

TOI-2046b

R-2183

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI2046_250913 · created 2026-08-13T23:01:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460931.89 +/- 0.2 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.066 +/- 0.046
Transit depth (Rp/Rs)^2	0.43 +/- 0.6 [%]
Orbital Inclination (inc)	82.56 +/- 2.87
Airmass coefficient 1 (a1)	0.6434 +/- 0.0085
Airmass coefficient 2 (a2)	0.316 +/- 0.093
Scatter in the residuals of the lightcurve fit is	1.32 %
Best Comparison Star	#1 - [435, 219]
Optimal Aperture	3.31
Optimal Annulus	6.5
Transit Duration (day)	0.024 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.066 ± 0.046 vs 0.1213 (deviation 1.17σ)
Duration fit vs published	0.024 d vs 0.1033 d (deviation 2.34σ)
Mid-time O–C	20.2 min
Depth SNR	1.4
Residual scatter	1.32 %

Light curve

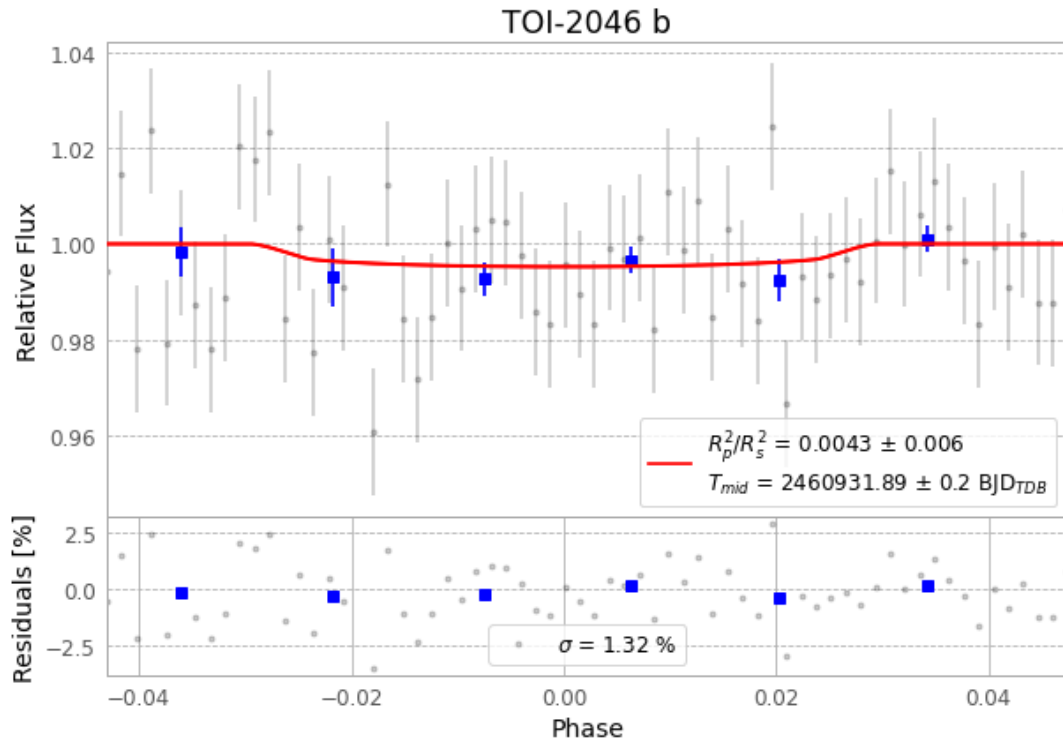


Figure 1 — FinalLightCurve_TOI-2046 b_2025-09-13.png (SHA-256 c28dcd5eb96802f2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [392, 257], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6240dde03692d668...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

66 FITS frames (43 MB), TOI2046250913075116.FITS → TOI2046250913110615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI2046-250913.log (977ffe21253b...), FinalLightCurve_TOI-2046 b_2025-09-13.png (c28dcd5eb968...), FinalLightCurve_TOI-2046 b_2025-09-13.csv (0220438f409a...)

Compute resources

Wall time 272.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 23:01Z.